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Communication device and power source device

Abstract

A communication device **100** of the present invention is placed at a place which is composed of a hole configured from a side wall **11** or/and a bottom and a cover **12** covering the hole, the communication device comprising: a plate member **101** obtaining heat of the cover; a heat transfer member **102** connected to the plate member to transfer the heat of the cover; a thermoelectric generation unit **103** in which a first junction is connected to the heat transfer member and a second junction is connected to the side wall, the bottom or liquid existing in the hole; and a communication unit **104** performing communication using electric power generated by the thermoelectric generation unit.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) The present application is based on and claims the benefit of priority from Japanese Patent Application No. 2021-056242 filed on Mar. 29, 2021, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a communication device provided with a maintenance hole or a hand hole and the like, and a power source device mountable on the communication device.

BACKGROUND

(3) There have been devices for detecting abnormality by obtaining environmental information of maintenance holes and hand holes. Also, a technique for collecting, storing, and utilizing various data by transmitting the data collected from afar using low-power and long-distance wireless communication scheme such as LPWA (Low Power Wide Area) and the like has been known.

PRIOR ART LITERATURE

Patent Literature

(4) Patent Literature 1: JP2006-233436 Patent Literature 2: JP2015-12431

SUMMARY OF THE INVENTION

(5) In case that environmental information of the maintenance holes and hand holes is detected by sensors and transmitted using wireless communication, a power source will be necessary for operating a communication device. However, the maintenance holes and the hand holes are usually covered by metal covers and sunlight does not reach into the maintenance holes and hand holes, and thus it is difficult to generate electric power by sunlight. Also, if a primary battery is used, it will be necessary to periodically replace the battery.

(6) The purpose of the present invention is to realize a communication device which does not need battery replacement.

(7) A communication device (**100**) according to one embodiment of the present application is a communication device placed at a place which is composed of a hole configured from a side wall (**11**) or/and a bottom and a cover (**12**) covering the hole, the communication device comprising: a plate member (**101**) obtaining heat of the cover; a heat transfer member (**102**) connected to the plate member to transfer the heat of the cover; a thermoelectric generation unit (**103**) in which a first junction is connected to the heat transfer member and a second junction is connected to the side wall, the bottom or liquid existing in the hole; a communication unit (**104**) performing communication using electric power generated by the thermoelectric generation unit. Here, the wording “heat” includes not only a case in which a temperature of the cover is higher than a temperature of the side wall, but also a case in which a temperature of the cover is lower than a temperature of the side wall, that is in which minus-evaluated heat is transferred from the cover to the side wall. Also, wording “obtain” means that heat is transferred to a plate member by transmission or conduction.

(8) The numbers in parentheses attached to the claim elements of the present disclosure described in this section and claims indicate the relationship between the present invention and the embodiments described below, and are not intended to limit the present invention.

Advantageous Effects of Inventions

(9) According to the present invention, it is possible to realize a communication device which does not need battery replacement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side view of a communication device in the present embodiment;
- (2) FIG. 2 is a front view of the communication device in the present embodiment;
- (3) FIG. 3 is a diagram illustrating a placing condition of plate member of the communication device in the present embodiment;
- (4) FIG. 4 is a diagram illustrating an operation of the communication device in daytime in present embodiment;
- (5) FIG. 5 is a diagram illustrating an operation of the communication device in night time in present embodiment; and
- (6) FIG. 6 is a diagram illustrating a placing condition of the communication device in the present embodiment.

DETAILED DESCRIPTION

- (7) Embodiments of the present invention will now be described with reference to the drawings.
- (8) Note that the present invention indicates any inventions in claims and not limited to the embodiments below. Further, at least words within the double quotation means words and phrases described in the claims and is not limited to the following embodiments.
- (9) Configurations and methods described in dependent claims of claims should be interpreted as arbitrary configurations and arbitrary methods in the invention of independent claims. Configurations and methods of following embodiments corresponding to configurations and methods described in dependent claims, and configurations and methods described only in the following embodiments without descriptions in claims should be interpreted as arbitrary configurations and arbitrary methods in the present invention. In a case that the scope of claims is broader than descriptions of the embodiments, configurations and methods described in the following embodiments are just examples of configurations and methods of the present invention, which should be interpreted as arbitrary configurations and arbitrary methods in the present invention. In any cases, essential configurations and methods of the present invention should be interpreted based on independent claims.
- (10) Any effects described in embodiments are effects obtained when a configuration of an embodiment as an example of this disclosure and are not necessarily an effect of the present invention.
- (11) When there are a plurality of embodiments, the configuration disclosed in each embodiment is not limited to each embodiment alone, and may be combined across the embodiments. For example, the configuration disclosed in one embodiment may be combined with another embodiment. Further, the disclosed configurations may be collected and combined in each of the plurality of embodiments.
- (12) The problem described in this disclosure is not a publicly known problem, but person including the inventor has independently found out, and is a fact that affirms the inventive step of the invention together with the configuration and method of the present disclosure.

Embodiment 1

- (13) 1. Places where a Communication Device and a Power Source Device of the Present Embodiment are Placed
- (14) In order for workers to access from ground for maintain (i.e. inspection, repairing, cleaning, ventilation, and the like) underground sewer, culvert, water supply, buried electrical and communication cables and buried gas pipe, vertical holes are provided with ground. Such a vertical

hole is called as maintenance hole or hand hole. Otherwise, in order to drain water on streets, gutters are provided with at least one side of streets.

(15) The maintenance hole, the hand hole, and the gutter (corresponding to “hole”) are composed of a side wall **11**. The side wall **11** is made from, for example, concrete or soil.

(16) Also, the maintenance hole, the hand hole, or the gutter is covered by a metal cover **12** having circle or rectangular shape so that people or object would not accidentally fall down.

(17) A communication device **100** and a power source device **150** of the present embodiment are placed at a space composing of the “hole” and a cover **12**.

(18) 2. Configuration of the Communication Device and the Power Source Device

(19) Referring to FIG. **1** and FIG. **2**, the configuration of the communication device **100** and the power source device **150** of the present embodiment will be described. FIG. **1** is a side view of the communication device **100** of the present embodiment, and FIG. **2** is a front view of the communication device **100** seeing from direction A of FIG. **1**. The communication device **100** comprises a plate member **101**, a heat transfer member **102**, a thermoelectric generation unit **103**, a communication unit **104**, a secondary battery **105**, and a primary battery **106**.

(20) The plate member **101** is a member having plate shape, which is sandwiched between a back surface of the cover **12** and an upper surface of the side wall **11** when the communication device **100** is placed at a space composed of the hole and the cover **12**.

(21) An upper surface of the plate member **101** contacts with the back surface of the cover **12** to obtain heat of the cover **12**. The plate member **101** is made from metal having high thermal conductivity, such as iron (including alloy), copper (including alloy), or aluminum. Otherwise, thermal conductive plastic or ceramics may be used.

(22) Elastically deformable leaf spring may be used as the plate member **101**. For example, as shown in FIG. **3(A)**, the plate member **101** has a shape in which a projecting part from the side wall **11** into the hole side is upwardly warping. Then, the hole is covered by the cover **12** to elastically deform the warping part by being pushed by the cover **12**, and as a result, as shown in FIG. **3(B)**, the back surface of the cover **12** is always applied stress from the plate member **101** as shown by the arrow. This configuration enables to keep a good contact between the plate member **101** and the cover **12**.

(23) Otherwise, the plate member **101** may have a shape in which, when the plate member **101** is provided so as to be sandwiched between the back surface of the cover **12** and the upper surface of the side wall **11**, an edge of the plate member **101** is warped toward the side wall **11** as shown in FIG. **3(C)** to generate stress pushing the thermoelectric generation unit **103** against the side of the side wall **11** via the heat transfer member **102**. This configuration enables to keep a good contact between the thermoelectric generation unit **103** and side wall **11**.

(24) It should be noted that any material having thermal insulation may be applied to the lower surface of the plate member **101**, that is, to a surface contacting the upper surface of the side wall **11**, or any thermal insulation may be sandwiched between the plate member **101** and the upper surface of the side wall **11**. This configuration enables to prevent heat of the cover **12** from being directly transferred to the side wall **11** via the plate member **101**. The plate member **101** is not necessary to be a plate shape. For example, it may be a shape which can contact the back surface of the cover, the “plate member” may be alternatively called as a cover contact member or may be read as a cover contact member.

(25) According to the plate member **101** of the present embodiment, there is a space between the cover **12** and the side wall **11** and an area directly contacting the cover **12** and the side wall **11** becomes smaller, which enables heat of the cover **12** to prevent from being directly transferred to the side wall **11**. Also, according to the plate member **101** of the present embodiment, there is a space between the cover **12** and the side wall **11** and radio wave used by the communication unit **104** described below is leaked from the space, which enables to improve communication quality to a base station device outside the hole.

(26) One end of the heat transfer member **102** is connected to the plate member **101**, and the other end is connected to the first junction of the thermoelectric generation unit **103** to transfer the heat of the cover **12** to the thermoelectric generation unit **103**.

(27) Any plate or rod shape member or heat pipe made from high thermal conductive material such as copper may be used as the heat transfer member **102**. Heat pipe is a member having a principle or structure in which heat is transferred by using transfer of hydraulic fluid inside. If the heat pipe is used, a plurality of heat pipes having circular cross section may be bundled.

(28) Otherwise, when placing the communication device **100**, it is possible to generate stress which pushes the thermoelectric generation unit **103** against the side of the side wall **11** by configuring or placing the heat transfer member **102** itself elastically deformable, which enables to keep a good contact between the thermoelectric generation unit **103** and the side wall **11**. It should be noted that the heat transfer member **102** and the plate member **101** may be configured to be as one member. In this case, this one member is the heat transfer member **102** and also the plate member **101**. Also, the thermoelectric generation unit **103** described below may be directly connected to the plate member **101**. In this case, the plate member **101** also has a function of the heat transfer member **102**.

(29) The thermoelectric generation unit **103** in which a first junction is connected to the heat transfer member **102** and a second junction is connected to the side wall **11** performs thermoelectric power generation using Seebeck effect. The first junction and the second junction include line or surface.

(30) The thermoelectric generation unit **103** is composed of a thermoelectric power generation element **107**, and thermally conductive adhesives **108** and **109**. The thermoelectric power generation element **107** is a power generation element using Seebeck effect, which is a power generation device in which one junction is contacted with high heat source and the other junction is contacted with low heat source to generate a potential difference to convert thermal energy into electric energy. The thermoelectric power generation element **107** may use power generation element made from, for example, Bismuth-tellurium (Bi—Te) based, Lead-tellurium (Pb—Te) based, silicon-germanium (Si—Ge) based material. The thermoelectric power generation element **107** may also use skeleton-type.

(31) Both surfaces of the thermoelectric power generation element **107** may be made from high thermal conductive material. Examples include copper, aluminum (after alumite treatment is preferable), stainless (e.g. SUS304), ceramics, or thermal conductive plastic.

(32) The thermally conductive adhesives **108** and **109** are provided on each surface of the thermoelectric power generation element **107**. For example, alfa-gel, silicone, or metallic adhesive may be used as the thermally conductive adhesives **108** and **109**. The thermally conductive adhesive **108** configures a first junction to be connected to the heat transfer member **102**. The thermally conductive adhesive **109** configures a second junction to be connected to the side wall **11**. In the present embodiment, a position of the second junction is an upper side of the side wall **11**, however, it may be near bottom of the hole. In this case, any heat transfer member such as a heat pipe may be provided between the thermally conductive adhesive **109** and the side wall **11**. In this case, the heat pipe also configures the thermoelectric generation unit **103**.

(33) In the present embodiment, the thermoelectric generation unit **103** is connected to the heat transfer member **102** and the side wall **11** via the thermally conductive adhesives **108** and **109**, however, the thermoelectric power generation element **107** may be directly connected to the heat transfer member **102** and the side wall **11** without the thermally conductive adhesives **108** and **109**. Instead of the thermally conductive adhesives **108** and **109**, any thermal conductive material having no adhesivity may be used.

(34) The communication unit **104** performs communication using electric power generated by the thermoelectric generation unit **103**. Any wireless communication scheme can be used as the communication scheme of the communication unit **104**. For example, broadband cellular

communication called 3G, 4G, 5G or low-power and long-distance wireless communication scheme (LPWA (Low Power Wide Area)) scheme which enables lower power consumption and long-distance communication may be used. LPWA scheme is a communication scheme which mainly uses 800/900 MHz band called Sub-Giga band, which includes, but is not limited to, eMTC (enhanced Machine Type Communication) proposed by 3GPP (Third Generation Partnership Project), NB-IOT (Narrow Band Internet of Things) optimized to perform small data communication, SIGFOX™ developed by Sigfox, or LoRa™ developed by Semtech. PARCA™ which features broadcast typed bidirectional communication proposed by the present applicant may be used. Otherwise, short-distance wireless communication scheme such as Wi-Fi™, ZigBee™, Bluetooth™, Bluetooth Low Energy (BLE), FeliCa™, NFC (Near Field Communication), or RFID may be used.

(35) The communication unit **104** may be configured to be supplied electric power generated by the thermoelectric generation unit **103** directly from the thermoelectric generation unit **103**, however, it may be alternatively configured to be supplied electric power, which is generated by the thermoelectric generation unit **103** and is stored at the secondary battery **105**, from the secondary battery **105**.

(36) Furthermore, the communication unit **104** may be supplied electric power from the primary battery **106**. Especially, when temperature difference between the cover **12** and the side wall **11** is lower than a predetermined threshold, enough electric power will not be supplied from the thermoelectric generation unit **103** and then electric power stored at the secondary battery **105** becomes insufficient. Only in such a case, the communication unit **104** may be supplied electric power from the primary battery **106** as a backup battery.

(37) The communication unit **104** transmits an output signal of a sensor obtaining environmental information. For example, the communication unit **104** transmits the output signal of the sensor by providing an input terminal of the sensor with the communication device **100** and connecting the sensor to the input terminal.

(38) Any sensor for obtaining necessary environmental information may be used. Examples include a temperature sensor for measuring temperature in the maintenance hole, a humidity sensor for measuring humidity in the maintenance hole. Other examples include a gas sensor for detecting gas in the maintenance hole, a corrosion potential sensor (voltmeter or ammeter) for measuring corrosion potential of the pipe in the maintenance hole.

(39) The communication unit **104** may comprise a RF circuit, a power circuit, or a sensing circuit.

(40) An antenna **110** is connected to the communication unit **104** to transmit the output signal output from the communication unit **104**. The antenna **110** may be an inverted-F board antenna or a whip antenna. It is desirable to provide the antenna **110** to be separated from the heat transfer member **102**. This enables to improve antenna sensitivity. Also, the antenna **110** may be grounded to the cover **12**. This enables to improve antenna sensitivity.

(41) The power source device **150** of the present embodiment comprises the plate member **101**, the heat transfer member **102**, and the thermoelectric generation unit **103**. The power source device **150** may further comprises the secondary battery **105** and the primary battery **106**.

(42) 3. Operation of the Communication Device and the Power Source Device

(43) Referring to FIG. 4 and FIG. 5, operations of the communication device **100** and the power source device **150** of the present embodiment will be described. FIG. 4 shows a condition in daytime, and FIG. 5 shows a condition in night time.

(44) As shown in FIG. 4, since the temperature of the cover **12** rises by receiving radiant heat of the sun in daytime, the temperature of the cover **12** is higher than the temperature of the side wall **11**. Therefore, the heat is transferred to the side wall **11** via the plate member **101**, the heat transfer member **102**, and the thermoelectric generation unit **103** in order. The temperature difference (i.e. heat flux) between the cover **12** and the side wall **11** at this time enables the thermoelectric generation unit **103** to generate electric power.

(45) On the other hand, as shown in FIG. 5, since the temperature of the cover **12** decreases because of the drop in outside temperature in night time, the temperature of the side wall **11** is higher than the temperature of the cover **12**. Therefore, the heat is transferred to the cover **12** via the side wall **11**, the thermoelectric generation unit **103**, the heat transfer member **102** in order. The temperature difference (i.e. heat flux) between the cover **12** and the side wall **11** at this time enables the thermoelectric generation unit **103** to generate electric power.

(46) However, in cases as shown in FIG. 4 and FIG. 5, since the direction of current generated by the thermoelectric generation unit **103** is reversed, the thermoelectric generation unit **103** can be regarded as an alternator with a long period. Therefore, by providing a rectifier circuit (charge circuit) not shown in figures with the thermoelectric generation unit **103**, it can convert from alternating current into direct current.

(47) It should be noted that the present embodiment was described on the premise of the situation shown in FIG. 4, however, in case of a situation shown in FIG. 5, the heat transferred direction is read as reverse direction. Otherwise, in case of the situation shown in FIG. 5, it should be interpreted that minus-evaluated heat is transferred from the cover **12** to the side wall **11**.

(48) 4. Other Configurations

(49) FIG. 6 is a diagram of the maintenance hole seen from above without the cover **12**. The plate member **101** of the communication device **100** of the present embodiment is placed so as to be put on the upper surface of the side wall **11**. In addition, dummy spacers **13** having the same thick as the plate member **101** may be provided with other area of the side wall **11**.

(50) Since the cover **12** would not directly contact with the side wall **11** by providing dummy spacers **13**, the heat of the cover **12** is not directly transferred to the side wall **11**. Therefore, the temperature difference between the cover **12** and the side wall **11** can be obtained. Also, since there is a space between the cover **12** and the side wall **11** by providing the dummy spacers **13**, the radio wave leaking from the space increases. As a result, it is possible to improve communication quality to the base station device outside the hole.

Embodiment 2

(51) In Embodiment 1, the second junction of the thermoelectric generation unit **103** is connected to the side wall **11**, however, in the present embodiment, a configuration in which the second junction is connected to other than the side wall **11** will be described.

(52) For example, in case that the hole is composed of the side wall **11** and the bottom, the second junction may be connected to the bottom of the hole. Otherwise, in case that there is liquid such as water in the hole, the second junction is provided with the liquid surface or in the liquid to be connected to the liquid. Alternatively, it may be connected to a gas pipe or a water pipe provided with the hole. In any cases, as described in Embodiment 1, the thermoelectric generation unit **103** and the second junction may be connected using the heat transfer member such as the heat pipe and the like. In this case, the heat pipe also configures the thermoelectric generation unit **103**.

(53) That is, by connecting the second junction to any member or medium which is not contacted with the cover **12**, it can use the temperature difference between the cover **12**.

(54) (Recapitulation)

(55) The Features of the communication device and the power source device in each of the embodiments of the present invention have been described above.

(56) Since the terms used in each embodiment are examples, the terms may be replaced with terms that are synonymous or include synonymous functions.

(57) The terms such as first, second, to N-th (where N is an integer) used in each of the embodiments and in the claims are used to distinguish two or more configurations and methods of the same kind and are not intended to limit the order or superiority.

INDUSTRIAL APPLICABILITY

(58) The communication device and the power source device of the present embodiment may be

used not only for the maintenance hole, hand hole or gutter, but also any places where temperature difference can be maintained.

Claims

1. A communication device placed at a place which is composed of a material defining a hole, and a cover covering the hole, the communication device comprising: a cover contact member obtaining heat of the cover; a heat transfer member connected to the cover contact member to transfer heat of the cover, a thermoelectric generation unit in which a first junction is connected to the heat transfer member and a second junction is connected to a side wall of the material defining the hole; a communication unit performing communication using electric power generated by the thermoelectric generation unit; and a lateral push member configured to generate stress pushing the thermoelectric generation unit against the side wall of the material defining the hole wherein the lateral push member comprises a portion of the cover contact member and the heat transfer member.
2. The communication device according to claim 1, wherein the cover contact member is sandwiched between the cover and an upper surface of the side wall when the communication device is placed at the space.
3. The communication device according to claim 1, wherein the cover contact member is elastically deformed so as to apply stress to a back surface of the cover.
4. The communication device according to claim 1, wherein the heat transfer member is a plurality of heat pipe.
5. The communication device according to claim 1 further comprising: a secondary battery storing electric power generated by the thermoelectric generation unit; and wherein the communication unit performs communication using electric power supplied from the secondary battery.
6. The communication device according to claim 5 further comprising: a primary battery; and wherein in case that the electric power stored at the secondary battery is insufficient, the communication unit performs communication using electric power supplied from the primary battery.
7. The communication device according to claim 1, wherein the communication unit transmits an output signal of a sensor which obtains environmental information.
8. The communication device according to claim 1, wherein the thermoelectric generation unit has a rectifier circuit which converts alternating current into direct current.
9. A power source device placed at a place which is composed of a material defining a hole, and a cover covering the hole, the power source device comprising: a cover contact member obtaining heat of the cover; a heat transfer member connected to the cover contact member to transfer the heat of the cover; a thermoelectric generation unit in which a first junction is connected to the heat transfer member and a second junction is connected to a side wall of the material defining the hole, and a lateral push member configured to generated stress pushing the thermoelectric generation unit against the side wall of the material defining the hole; wherein the lateral push member comprises a portion of the cover contact member and the heat transfer member.
10. A communication device placed at a place which is composed of a material defining a hole, and a cover covering the hole, the communication device comprising: a cover contact member obtaining heat of the cover, connected to a surface of the cover; a heat transfer member connected to the cover contact member to transfer heat of the cover; a thermoelectric generation unit including a thermoelectric power generation element, a first thermally conductive adhesive on a first surface of the thermoelectric power generation element and a second thermally conductive adhesive on a second surface of the thermoelectric power generation element, wherein the first thermally conductive adhesive contacts with the heat transfer member, and the second thermally conductive adhesive contacts with a side wall of the of the material defining the hole; and a

communication unit performing communication using electric power generated by the thermoelectric generation unit a lateral push member configured to generate stress pushing the thermoelectric generation element against the side wall of the material defining the hole wherein the lateral push member comprises a portion of the cover contact member and the heat transfer member.

11. The communication device according to claim 1, further comprising: an antenna connected to the commutation unit to transmit an output signal output from the communication unit, wherein said antenna is grounded to the cover.

12. The communication device according to claim 10, further comprising: an antenna connected to the commutation unit to transmit an output signal output from the communication unit, wherein said antenna is grounded to the cover.
